# Multiplying and Dividing in Scientific Notation

*Lesson Plan*

## 1. Lesson Information

| | | | |
|---|---|---|---|
| Course | Grade 7 Accelerated (1205050) | Unit | 4 — Solving Problems with Rational Numbers |
| Lesson | Lessons 2–3 — Multiplying and Dividing in Scientific Notation | Date | _____ |
| Length | 53 minutes (one period) | Benchmark(s) | MA.8.NSO.1.5 |

## 2. Standards

**MA.8.NSO.1.5** Add, subtract, multiply and divide numbers expressed in scientific notation with procedural fluency.

**Must.** Multiply and divide numbers written in scientific notation with procedural fluency, and write every answer with a coefficient at least 1 and less than 10.

**Must not.** No within-2 limit applies here — that clarification is about addition and subtraction only, and the contrast with 4.01 is the point of the day. Significant digits are 4.04.

## 3. Mathematical Thinking and Reasoning Standards

*the evidence for each, from this period*

| MTR | What it asks | The evidence in this lesson |
|---|---|---|
| MA.K12.MTR.5.1 | Use patterns and structure to help understand and connect mathematical concepts. | Notes I — the product is regrouped before it is evaluated, so 'multiply the fronts, add the backs' is read off the structure rather than memorised as a rule. |
| MA.K12.MTR.3.1 | Complete tasks with mathematical fluency. | Boards 1, 3 and 7 — one move each, written and up on the cue; dividing in scientific notation was the lowest single figure on the last state report. |
| MA.K12.MTR.4.1 | Engage in discussions that reflect on the mathematical thinking of self and others. | Board 6 — when the room splits between $2 \times 10^{10}$ and $20 \times 10^{9}$, which are the same number, the sixty-second re-vote before the reveal. |

## 4. Learning Target and Essential Question

**Learning target.** I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

**Essential question.** Why must the powers of ten match to add, but not to multiply?

**Building on.** 4.01 — the coefficient rule and the finishing step; MA.7.NSO.1.1 and MA.8.NSO.1.3 (3.01–3.03) — the product and quotient of powers laws.

**Working toward.** 4.04 — real-world problems and significant digits; MA.912.NSO.1 — operations with rational exponents in Algebra 1.

## 5. Sequence

*read from the deck; never typed*

| Segment | Min | What the teacher does | What students do |
|---|---|---|---|
| Settle in; post the learning target | 1 | Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet. | Copy the target. |
| Warm-Up | 6 | Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once. | Four retrieval questions, alone, no notes. Then copy the new words. |
| Notes | 11 | Teach from the front. Expand once per law; say the OFF line out loud. | Copy the notes; answer the checks aloud. |
| Example 1 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Example 2 | 3 | Work the example; name the wrong boards before they appear. | Watch, then work the Your Turn alone. |
| Your Turn | 3 | Circulate; reveal and name what a wrong board most likely did. | Work it alone; boards up on the cue. |
| Whiteboards | 12 | One question at a time. Boards down until the cue, all up together; scan the back row first. | Answer on a board; hold it up on three. |
| Independent Set | 6 | Silent. Circulate and mark what you see; do not teach. | Six questions, written, alone. |
| IXL | 5 | Assign the day's IXL skills. | Start the skills; finish tonight. |
| **Total** | **53** | | |

## 6. Gradual Release

**I do.** The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

**We do.** Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

**You do.** The nine-question whiteboard round (12 minutes) and then the six-question independent set, silent and written.

## 7. Higher-Order Questions

| Question | DOK |
|---|---|
| Why do the powers of ten have to match before you add, but not before you multiply? | DOK 3 |
| Marcus's power of ten is right and Sara's coefficient is right. Which of the two is closer to the correct answer, and does 'closer' mean anything useful here? | DOK 3 |
| A quotient of two positive numbers comes out as $5.3 \times 10^{-3}$. What does the negative exponent tell you about how the two numbers compared? | DOK 2 |

## 8. Checks for Understanding

*each board, and what to do about it*

| # | What the board asks | Answer | If the board is wrong |
|---|---|---|---|
| 1 | $(3\times10^5)(3\times10^6)$ | ***9 × 10¹¹*** | 9 × 10³⁰ — the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; 6 × 10¹¹ — the coefficients were added [8.NSO.1.5: the operation is multiplication] |
| 2 | A printer makes 4 × 10³ copies per hour. It runs for 2.5 × 10² hours. Paper costs 3 cents a sheet. How many copies does it make? | ***1 × 10⁶*** | anything built from 3 cents — the unneeded figure was used [the question asks for a count, not a cost]; 10 × 10⁵ — right value, coefficient not less than 10 [B1G-M 8.NSO.1.5 misconception] |
| 3 | $\frac{9.6\times10^8}{1.2\times10^3}$ | ***8 × 10⁵*** | 8 × 10¹¹ — the exponents were added [8.NSO.1.5: division subtracts the exponents]; 8.4 × 10⁵ — the coefficients were subtracted instead of divided [8.NSO.1.5] |
| 4 | $\frac{3.28\times10^7}{4\times10^{10}}$ | ***8.2 × 10⁻⁴*** | 0.82 × 10⁻³ — right value, but 0.82 is not at least 1 [8.NSO.1.5: the coefficient satisfies 1 ≤ a < 10]; 8.2 × 10⁻³ — the point moved but the exponent did not |
| 5 | $(4 \times 10^7)(2 \times 10^9)$. Which is the product? | ***C — 8 × 10¹⁶*** | (A) the exponents were subtracted, 7 − 9 [8.NSO.1.5: multiplication adds the exponents]; (B) the exponents were multiplied [B1G-M 8.NSO.1.5 named misconception: $(2 \times 10^4)(3 \times 10^5) = 6 \times 10^{20}$]; (D) the coefficients were added instead of multiplied [8.NSO.1.5: the operation is multiplication] |
| 6 | $(5 \times 10^6)(4 \times 10^3)$. Which is the product, written in scientific notation? | ***D — 2 × 10¹⁰*** | (A) the coefficient was repaired but the exponent was not raised [8.NSO.1.5: point one place left, exponent up one]; (B) equal in value, but 20 is not less than 10 [B1G-M 8.NSO.1.5 named misconception: leaving 12 × 10⁹ instead of 1.2 × 10¹⁰]; (C) the exponents were multiplied and the coefficient was left over 10 [8.NSO.1.5] |
| 7 | $\frac{7\times10^9}{7\times10^{-6}}$ | ***1 × 10¹⁵*** | 1 × 10³ — the negative sign on the bottom exponent was dropped [8.NSO.1.5: subtracting a negative adds]; 1 × 10⁻¹⁵ — bottom minus top [8.NSO.1.5: top minus bottom] |
| 8 | $(8.4 \times 10^7) \div (n \times 10^2) = 4.2 \times 10^5$. What is the value of n? | ***n = 2*** | 4.2 — the coefficients were subtracted, 8.4 − 4.2 [8.NSO.1.5: the coefficients divide]; 0.5 — 4.2 ÷ 8.4 instead of 8.4 ÷ 4.2 |
| 9 | For $(7.2 \times 10^{11}) \div (2.4 \times 10^3)$, Marcus writes 4.8 × 10⁸ and Sara writes 3 × 10¹⁴. Name what each one did wrong, and give the correct quotient. | ***3 × 10⁸*** | 'they both got it wrong' — neither error is named [the item asks what each one did]; 4.8 × 10⁸ — Marcus's error repeated |

**MTR.4.1 — the one peer move.** When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

## 9. Differentiation

**ESE / IEP.** Give a two-column organiser — COEFFICIENTS on the left, POWERS OF TEN on the right — and have the student split every expression into the two columns before computing anything. The regrouping step in Notes I becomes a physical act, and the two laws are then applied one column at a time.

**ELL.** Product and quotient are the two nouns, multiply and divide the two verbs. Write them in pairs on the board — PRODUCT goes with MULTIPLY and ADD the exponents, QUOTIENT goes with DIVIDE and SUBTRACT the exponents — and read each board's operation aloud before the boards go up.

**Enrichment.** Ask for $(4 \times 10^{5})^{2}$ without a calculator ($1.6 \times 10^{11}$), then for the two numbers in scientific notation whose product is $1.2 \times 10^{7}$ and whose quotient is $1.2 \times 10^{1}$ ($1.2 \times 10^{4}$ and $1 \times 10^{3}$).

## 10. Closure, Vocabulary, Materials, Practice

**Closure.** Board 9 is written work and it is the exit evidence: the student has to name Marcus's error and Sara's error separately and give $3 \times 10^{8}$. Read the boards, not the papers.

**Vocabulary.** **product of powers law**; **quotient of powers law**; **regrouping a product**

**Materials.** Whiteboards and markers. Calculators are allowed on everything; a calculator is the fastest way to settle whether $8 \times 10^{16}$ and $8 \times 10^{63}$ could possibly both be the answer to whiteboard 5.

**Practice.** The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Multiply numbers written in scientific notation — YZU; Divide numbers written in scientific notation — SGT.